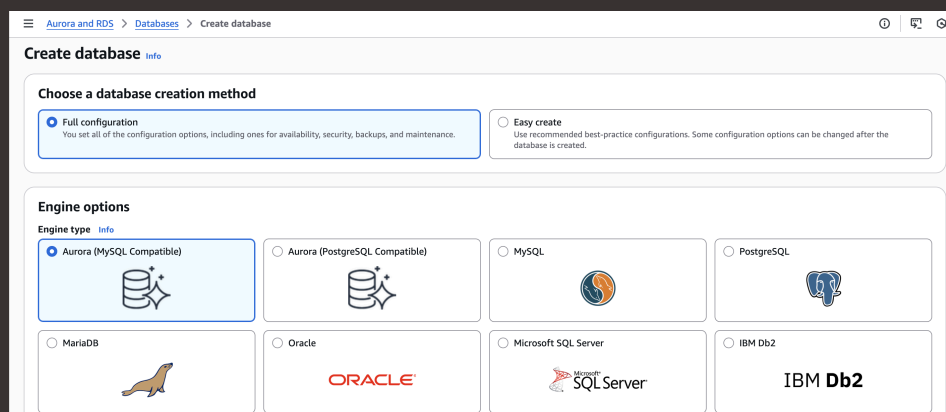




Connect a Web App to Amazon Aurora

CH

Arpita Chowdhury





Introducing Today's Project!

What is Amazon Aurora?

Amazon Aurora is a fully managed, MySQL/PostgreSQL-compatible relational database service designed for high performance and high availability using a cluster-based distributed storage system. It is useful because it provides up to 5× faster performance than standard MySQL, offers seamless read scaling, automatically replicates data across multiple Availability Zones, supports automatic failover, and reduces operational overhead making it ideal for modern, highly available applications.

How I used Amazon Aurora in this project

In today's project, I used Amazon Aurora to create a database and connect to an EC2 instance.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was to create a default VPC from scratch to use in this project.



Arpita Chowdhury

NextWork Student

nextwork.org

This project took me...

This project took me 2 hours.



In the first part of my project...

Creating an Aurora Cluster

A relational database is a structured database that organizes data into related tables and uses SQL to manage and query that data.

Aurora is a good choice when you need a large-scale, production-ready relational database with top-tier performance and high uptime.

The screenshot shows the 'Create database' page in the AWS Management Console. The breadcrumb navigation at the top reads 'Aurora and RDS > Databases > Create database'. The page title is 'Create database' with an 'Info' link. Under the 'Choose a database creation method' section, there are two options: 'Full configuration' (selected with a blue radio button) and 'Easy create'. Below this, the 'Engine options' section is displayed. It features a grid of database engine choices, each with a radio button, a name, and a logo. The first row includes 'Aurora (MySQL Compatible)' (selected), 'Aurora (PostgreSQL Compatible)', 'MySQL', and 'PostgreSQL'. The second row includes 'MariaDB', 'Oracle', 'Microsoft SQL Server', and 'IBM Db2'.

Choose a database creation method	
☒ Full configuration You set all of the configuration options, including ones for availability, security, backups, and maintenance.	☐ Easy create Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options			
Engine type Info			
☒ Aurora (MySQL Compatible) 	☐ Aurora (PostgreSQL Compatible) 	☐ MySQL 	☐ PostgreSQL
☐ MariaDB 	☐ Oracle 	☐ Microsoft SQL Server 	☐ IBM Db2



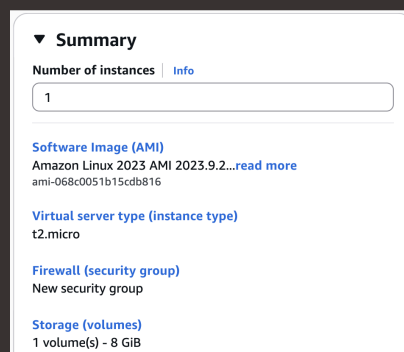
Halfway through I stopped!

I stopped creating my Aurora database because I need to create an EC2 instance.

Features of my EC2 instance

I created a new key pair for my EC2 instance because AWS requires a key pair to securely authenticate SSH access. The public key is stored on the EC2 instance, and the private key (which only I have) is used to prove my identity and log in. Without this key pair, I wouldn't be able to access or manage the server.

When I created my EC2 instance, I took particular note of Public IPv4 DNS and key pair name as both are crucial for accessing EC2.





Then I could finish setting up my database

Connectivity Info

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

EC2 instance Info
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

nextwork-ec2-instance-web-server

Some VPC settings can't be changed when a compute resource is added
Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Network type Info
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Aurora uses a cluster architecture to provide high availability, automatic failover, read scaling, and a shared distributed storage layer that is independent of the compute instances.



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